

Project Design Phase Solution Architecture

Date	01 August 2026
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Project Name	Intergalactic Space Travel Request System (ISTR)
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. For ISTR, its goals are to:

- Find the best tech solution (ServiceNow custom table + Flow Designer) to solve the interplanetary mission-approval problem.
- Describe the structure, characteristics, behavior, and other aspects of ISTR to project stakeholders (Requester, Manager, Finance Officer, Mission Control, Admin).
- Define features, development phases, and solution requirements for the ISTR platform.
- Provide specifications according to which ISTR is defined, managed, and delivered across Dev, Test, and Prod instances.

1. Solution Architecture Diagram (Data Flow View):

This view shows how a mission request moves through the platform end to end: from submission, through role-based approval routing, SLA tracking, and audit logging, to launch and stakeholder notification.

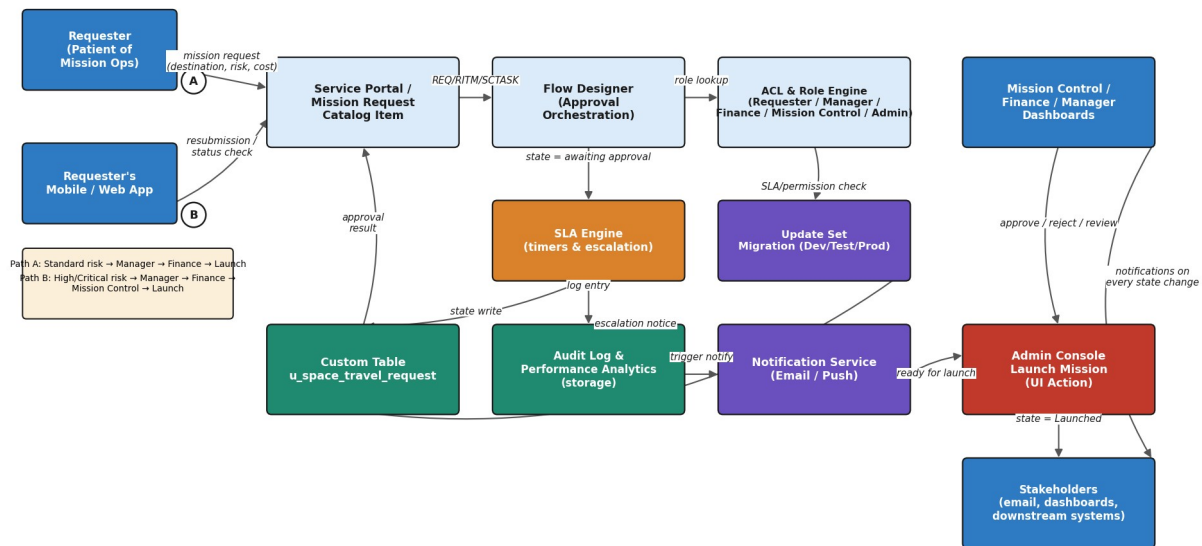


Figure 1: Architecture and data flow of the ISTR mission-approval application

2. Layered (3-Tier) Component Architecture:

This view organizes the same system by architectural layer — Presentation, Application/Logic, and Data — showing which ServiceNow components live in each tier and how they connect vertically.

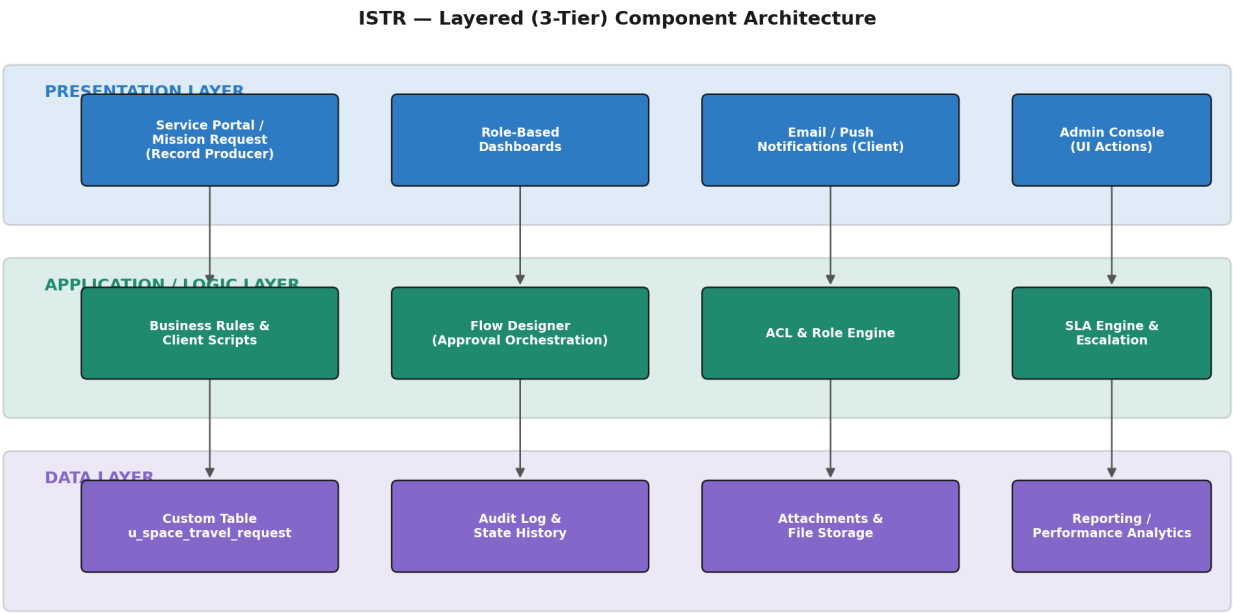


Figure 2: ISTR 3-tier component architecture

3. Mission Approval Sequence (Swimlane):

This view traces a single mission request over time across the five actor roles, showing exactly which step each stakeholder performs and where the flow branches for High/Critical risk missions.

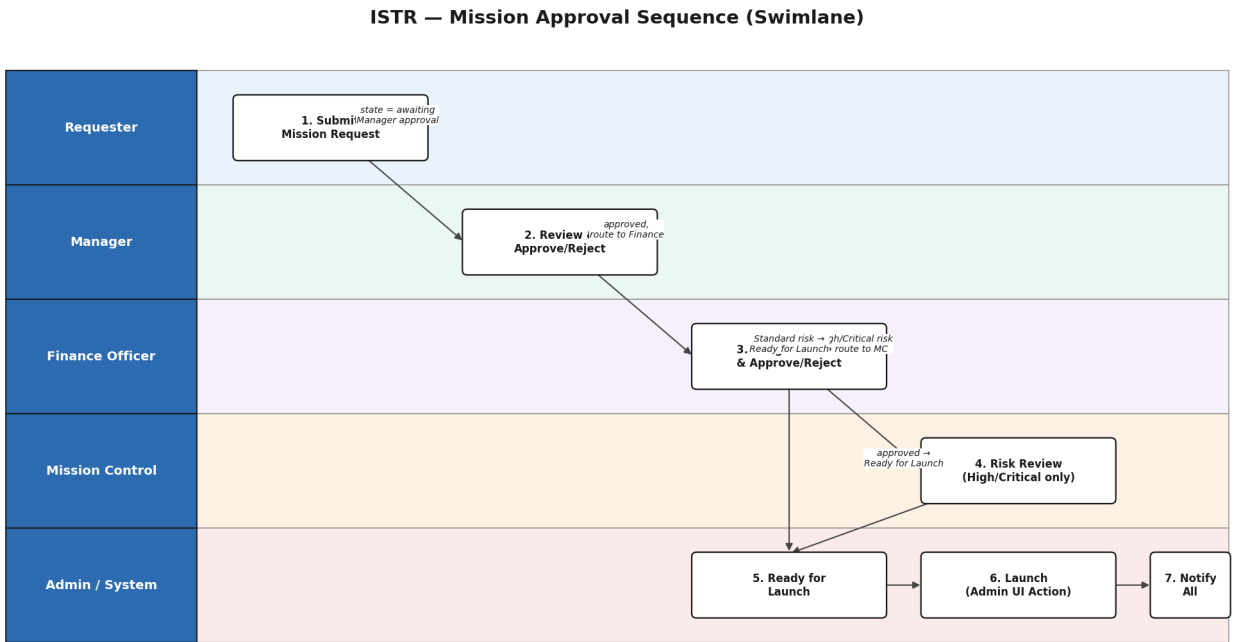


Figure 3: End-to-end approval sequence across Requester, Manager, Finance, Mission Control, and Admin

4. Deployment & Environment Flow:

This view shows how ISTR is built and promoted across ServiceNow instances — Dev, Test, and Prod — via Update Sets, and where external services (email notifications, MID Server/API integrations) connect in.

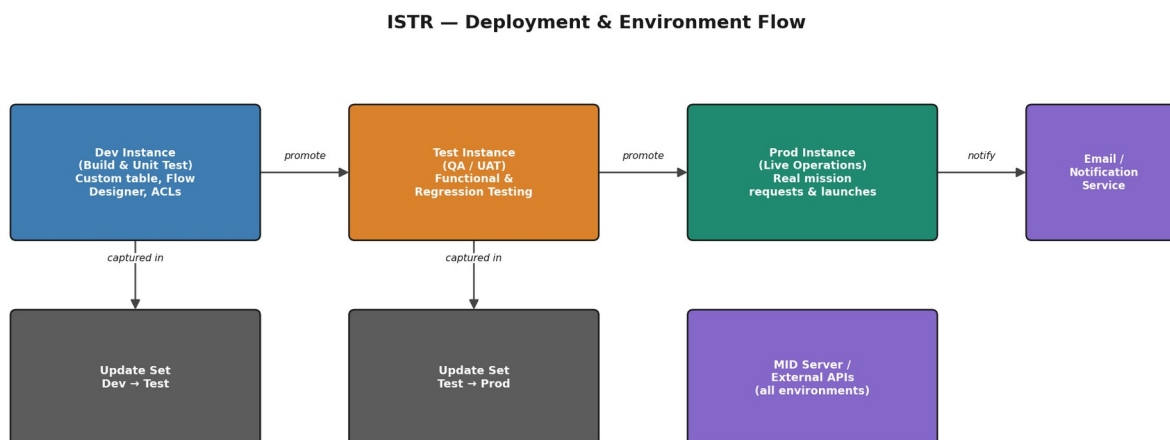


Figure 4: ISTR deployment pipeline across Dev, Test, and Prod instances

References:

1. <https://developer.servicenow.com/dev.do> (ServiceNow Developer Documentation — Flow Designer & Platform Architecture)
2. <https://www.servicenow.com/products/application-development.html>